College Timetable Optimizer

PROTOTYPE

CLASH-FREE

Prototype summary: what has been built so far

A web app that builds **clash-free college timetables automatically** instead of by hand, with a teacher-facing experience on top. It runs entirely in the browser using invented sample data: no server, database or login is needed.

Hackathon	Programme	Owner	Date
Scheduling Optimizer: Hackathon 1	**Punjab Jobs AI Bootcamp**	**Purvika**	**21 October 2026**

SESSIONS PLACED	CLASHES	AUTOMATED TESTS	SCREENS
74 / 74	**0**	**33 / 33**	**5**
standard sample college	teacher, room and section	passing	plus class-details dialog

01 The demo scenario

Who you are

Signed in as a sample teacher, **Dr. Amandeep Singh**, Associate Professor, Computer Science and Engineering, at an invented college, *Sutlej Valley Institute of Technology*.

The sample college

4 sections, 8 teachers, 6 rooms (lecture rooms and labs), 15 subjects, 74 weekly sessions. Monday–Friday with 8 periods, half-day Saturday, lunch break after period 4.

02 Demo walkthrough (about 3 minutes)

1. **Dashboard**: the teacher's day at a glance: next class, today's schedule, weekly load.
2. **My Timetable**: switch Week and Day views; open a lab, try a clashing move (refused with reasons), then a green suggestion (saved).
3. **My Availability**: mark a period unavailable and save; refresh to show it persists.
4. **Generate Timetable**: standard data gives 74/74 and 0 clashes; switch to section or room views; *Use as my timetable*.
5. **Stress test**: 4 unplaced sessions, each with a clear reason.
6. **Profile**: *Reset demo data* restores the original sample.

03 Screens and features

SCREEN	WHAT IT DOES
1 Dashboard	Greeting, date and department; **next class** (subject, time, room, section, countdown); today's schedule with free periods and lunch; upcoming classes this week; weekly load, free periods and sections taught; per-day load chart; clash-status line that warns if the timetable conflicts with saved availability.
2 My Timetable	**Week view** (Mon–Sat, 2-period labs, hatched lunch break, free periods, today highlighted) and a **Day view** with day navigation. Colour-coded subject cards that always show the subject code too.
2a Class details and move	Click any class to see subject, section, room, teacher, day, time and duration. **Move class** suggests only clash-free slots; a move that would clash is refused with the exact reason, such as "Room booked twice".
3 My Availability	Tap periods to mark available or unavailable, or a whole day off. **Saved in the browser** and kept after a refresh. Refuses to save if fewer periods remain than you teach; warns when a choice conflicts with current classes.
4 Generate Timetable	Runs the scheduler on the whole sample college. Shows scheduled and unplaced counts, **teacher, room and section clash counts**, validation status, and the timetable by teacher, section or room. Regenerate with a new seed; **Use as my timetable** applies a validated result after confirmation.
5 Profile	Teacher details, teaching assignments, sample-data notice, and **Reset demo data** with a confirmation.

04 Technology and how to run it

Stack

- Next.js 15 (App Router), React 19, TypeScript
- Tailwind CSS 4; inline SVG icons; native dialogs
- No other runtime packages
- Browser localStorage for saved data
- Node's built-in test runner

No Docker, Python, database, backend server, API keys or environment secrets needed.

Run on Windows (Node 22.18+)

Open a terminal in the repository root:

```
cd "C:\Users\anand\Desktop\timetable, hackathon 1"
```

```
# first time only
npm run setup
npm run dev
```

Then open **http://localhost:3000**.

05 How timetable generation works

1 Expand
Each course requirement becomes its weekly sessions.

2 Order
Hardest first: labs, busy teachers, few suitable rooms.

3 Place
Best valid slot per session using a penalty score.

4 Retry
Failed sessions go first; the best attempt is kept.

✓ Validate
An independent checker re-tests every rule.

Hard rules (never broken)

- No teacher, room or section double-booked
- Teacher availability respected
- Room type and capacity fit the class
- Labs are one 2-period block, never across lunch
- Each teacher's daily teaching limit respected

Soft preferences (scored)

- Spread a subject across the week
- Balance daily load for sections and teachers
- Labs in the afternoon where possible
- Avoid first and last periods
- Same seed always gives the same timetable

✓ **Standard semester:** all 74 sessions placed. The independent validator found 0 teacher, 0 room and 0 section clashes, and every availability, room and break rule holds.

! **Over-constrained stress test:** 76 of 80 placed, still 0 clashes. The 4 unplaced sessions are reported with reasons, never dropped silently:

- Compiler Design (×2): no lecture room seats 85 students; the largest holds 72
- IoT Workshop lab: the college has no workshop room
- Cloud Computing (1 of 3): the visiting teacher is available for only 2 periods

Sample output: Dr. Amandeep Singh's week (excerpt)

Period	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
P1 9:00 AM	Free	Free	Free	Free	Free	**CS301** Data Structures · 3B
P2 9:50 AM	Free	**CS501** Algorithms · 5A	**CS301** Data Structures · 3A	Free	**CS501** Algorithms · 5A	**CS301** Data Structures · 3A
P3 10:40 AM	**CS501** Algorithms · 5A	Free	**CS301** Data Structures · 3B	Free	Free	Free
Lunch break · 12:20 PM – 1:00 PM						
P6 1:50 PM	Free	**CS351 · LAB 2P** DS Lab · 3B Networks Lab	Free	Free	**CS351 · LAB 2P** DS Lab · 3A Programming Lab 1	
P7 2:40 PM	**CS301** Data Structures · 3A		Free	Free		

Excerpt of a generated result (periods 4, 5 and 8 omitted). Labs span two periods; hatched cells are outside Saturday's half day.

06 Quality and verification

Automated checks (all passing)

- ✓ **33 tests**: teacher, room and section clashes, lab overlaps, availability, capacity, room type, labs across lunch
- ✓ Infeasible cases reported as unplaced with reasons
- ✓ Same seed gives the same timetable; 25 seeds all clash-free
- ✓ Moves rejected on clash; suggested moves stay clash-free
- ✓ Corrupted or outdated saved data handled safely
- ✓ TypeScript check, ESLint and production build

Checked in the browser

- ✓ Dashboard, week and day views, class details
- ✓ Clashing move refused; valid move saved and kept after reload
- ✓ Availability saved and kept after reload; too-few-periods save blocked
- ✓ Standard and stress-test generation; applying a result
- ✓ Reset demo data; corrupted storage falls back to the sample
- ✓ Desktop and 375 px phone widths, no console errors

User experience

Loading placeholders, empty states, success messages and error and warning banners · keyboard operable with visible focus · labelled timetable cells for screen readers · confirmations for destructive actions · sidebar on desktop, bottom tab bar on phones · subject colour always paired with its code, never colour alone.

07 Limitations and next steps

Current limitations

- Sample data only; you are always the one sample teacher
- Data stays in one browser; not shared or backed up
- Whole-section classes only (no batches, electives, co-teaching)
- Fast heuristic generator, suited to small colleges
- English only; no PDF export, alerts or one-day changes

Planned (designed in `docs/`)

- Multi-college platform with secure data separation
- Roles: principal, coordinator, HOD, teacher, student
- Backend, database (Indian region) and optimisation solver
- Spreadsheet import; publishing with undo
- SMS and email alerts; Punjabi; offline student view